Katarina Petrović

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Education

University of Oxford (Expected graduation 06/2027) 09/2024 – Present
DPhil in Computer Science Oxford, United Kingdom
Supervised by DeepMind Chair of AI Prof. Michael Bronstein, Prof. İsmail Ceylan, and Dr. Alexander Tong

University of Oxford 09/2017 – 06/2021
MEng in Engineering Science Oxford, United Kingdom
Supervised by Prof. Xiaowen Dong and Prof. Jan-Peter Calliess

Mathematical Grammar School 09/2011 – 06/2017
High School Diploma Belgrade, Serbia
School for gifted students in Serbia

Publications

First Author of Global Transport Couplings for Classifier-Free Guided Flows **NeurIPS 2026 STODY**
Coupling strategy for improving classifier-free guidance and distillation into one- and few-step generators

First Author, Curly Flow Matching for Learning Non-gradient Field Dynamics **NeurIPS 2025**
Flow-based generative model that learns non-gradient field dynamics by solving Schrödinger bridge with non-zero drift

First Author, Temporal Graph Rewiring with Expander Graphs **ICML 2024 GRaM**
First method for rewiring continuous-time dynamic graphs using expander graphs

Author, EIT in Multilayered Spherical Quantum Dot with Hydrogenic Impurity **Elsevier Optical Materials**
Theoretical study of electromagnetically induced transparency in a multilayered quantum dot under varying conditions

Honours and Awards

ICLR 2025 MLGenX Workshop – Best Poster Award
Simon and June Li Scholarship – *Full-ride Oxford scholarship* awarded to *1-2* students p.a. *globally*
Romanian Master of Physics – *Bronze Medal* at an international physics competition held in Bucharest
Serbian National Physics Competition – *Ranked top 1* at a national physics competition

Academic

Teaching

Teaching Assistant, Geometric Deep Learning, University of Oxford Hilary Term, 2026
Internal Mentor, Group Design Project, University of Oxford Trinity Term, 2025
Teaching Assistant, Geometric Deep Learning, University of Oxford Hilary Term, 2025

Invited Talks

Flow Matching for Learning Cell Dynamics, In Silico #004, London, United Kingdom February 2026
Flow Matching & Drug Discovery Tutorial, EEML 2025 Summer School, Sarajevo, Bosnia & Herzegovina July 2025
Curly Flow Matching for Learning Non-gradient Field Dynamics, MBZUAI Workshop on Frontiers in Generative Modelling and Scientific AI, Abu Dhabi, UAE April 2025

Service

Peer Reviewer – ICLR 2027, NeurIPS 2026, ICLR 2026, ICLR 2025

Experience

Xaira Therapeutics, AI Scientist Intern, London, United Kingdom 08/2026 – Present
AITHYRA, Predoctoral Fellow, Vienna, Austria 01/2026 – Present
University of Pennsylvania, Visiting Student, Philadelphia, United States 08/2025 – 09/2025
Network Science Institute (Barabási Lab), Research Intern, Boston, United States 11/2023 – 03/2024
A&M Capital Europe, Private Equity Analyst, London, United Kingdom 05/2023 – 10/2023
The Carlyle Group, Private Equity Analyst, London, United Kingdom 07/2021 – 12/2022
Goldman Sachs, Investment Banking Summer Analyst, London, United Kingdom 07/2020 – 08/2020